HDOC DAY-62

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BATCH-15

Question

Write a C program to merge two sorted variable-length integer arrays using pointers and a function merge_sorted_arrays(). The merged array should maintain ascending order and preserve duplicate elements using the two-pointer technique.

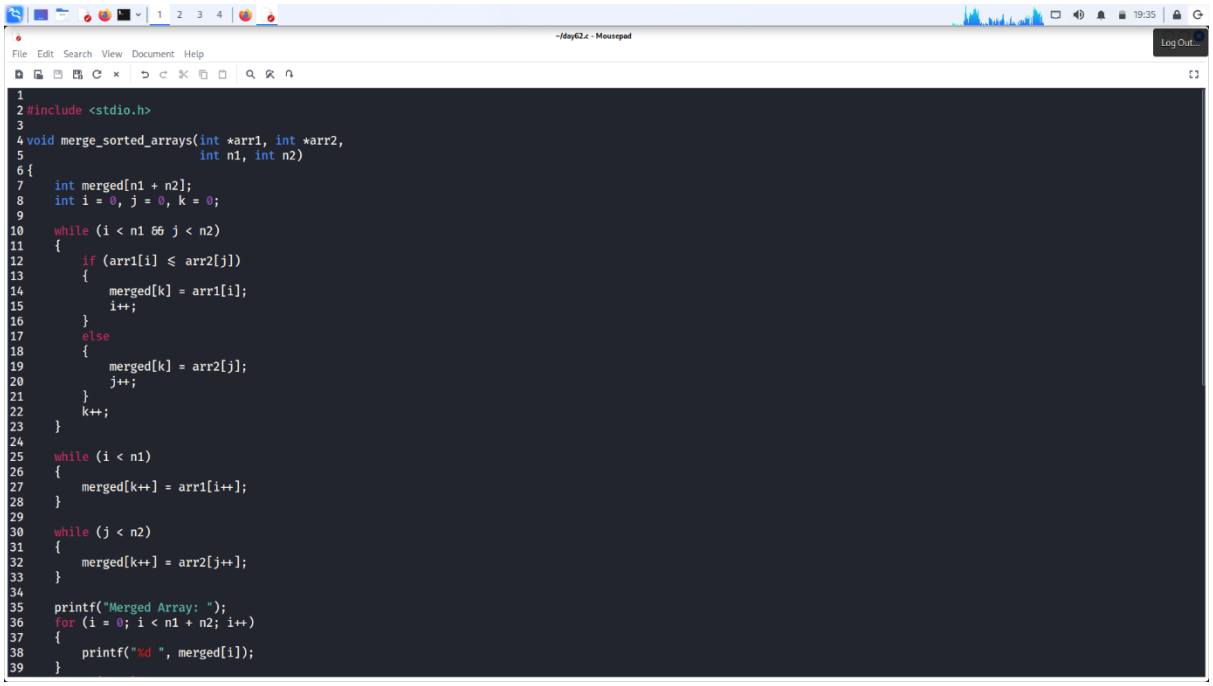
Aim

To write a C program to merge two sorted arrays into a third array in ascending order using pointers, variable-length arrays, and the two-pointer technique.

Algorithm

1. Start the program.
2. Read the sizes of two arrays, n1 and n2.
3. Declare two variable-length arrays, arr1[n1] and arr2[n2].
4. Read the elements of both arrays in ascending order.
5. Pass both arrays and their sizes to merge_sorted_arrays() using pointers.
6. Declare a third array merged[n1+n2].
7. Initialize three variables i=0, j=0, and k=0.
8. Compare the elements of both arrays and insert the smaller element into merged[].
9. Copy the remaining elements of either array into merged[].
10. Display the merged array in ascending order.
11. Stop the program.

>c code:



```c

#include <stdio.h>

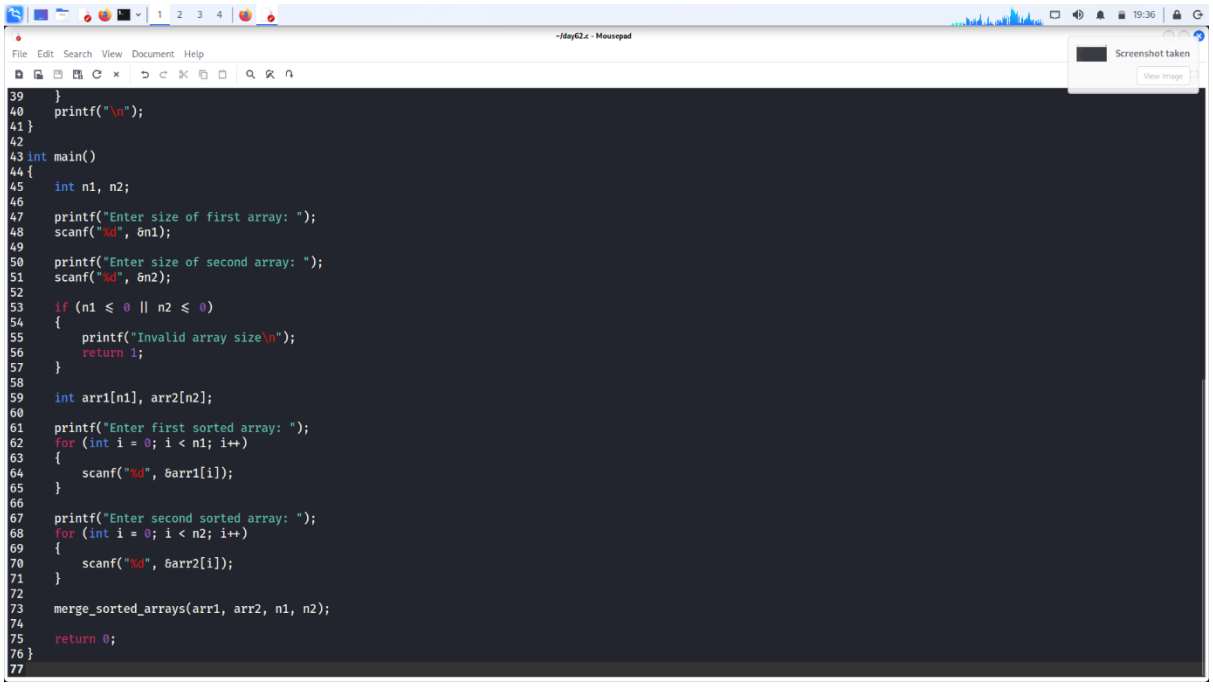
void merge_sorted_arrays(int *arr1, int *arr2,
                         int n1, int n2)
{
    int merged[n1 + n2];
    int i = 0, j = 0, k = 0;

    while (i < n1 && j < n2)
    {
        if (arr1[i] <= arr2[j])
        {
            merged[k] = arr1[i];
            i++;
        }
        else
        {
            merged[k] = arr2[j];
            j++;
        }
        k++;
    }

    while (i < n1)
    {
        merged[k++] = arr1[i++];
    }

    while (j < n2)
    {
        merged[k++] = arr2[j++];
    }

    printf("Merged Array: ");
    for (i = 0; i < n1 + n2; i++)
    {
        printf("%d ", merged[i]);
    }
```



```c
    }
    printf("\n");
}

int main()
{
    int n1, n2;

    printf("Enter size of first array: ");
    scanf("%d", &n1);

    printf("Enter size of second array: ");
    scanf("%d", &n2);

    if (n1 <= 0 || n2 <= 0)
    {
        printf("Invalid array size\n");
        return 1;
    }

    int arr1[n1], arr2[n2];

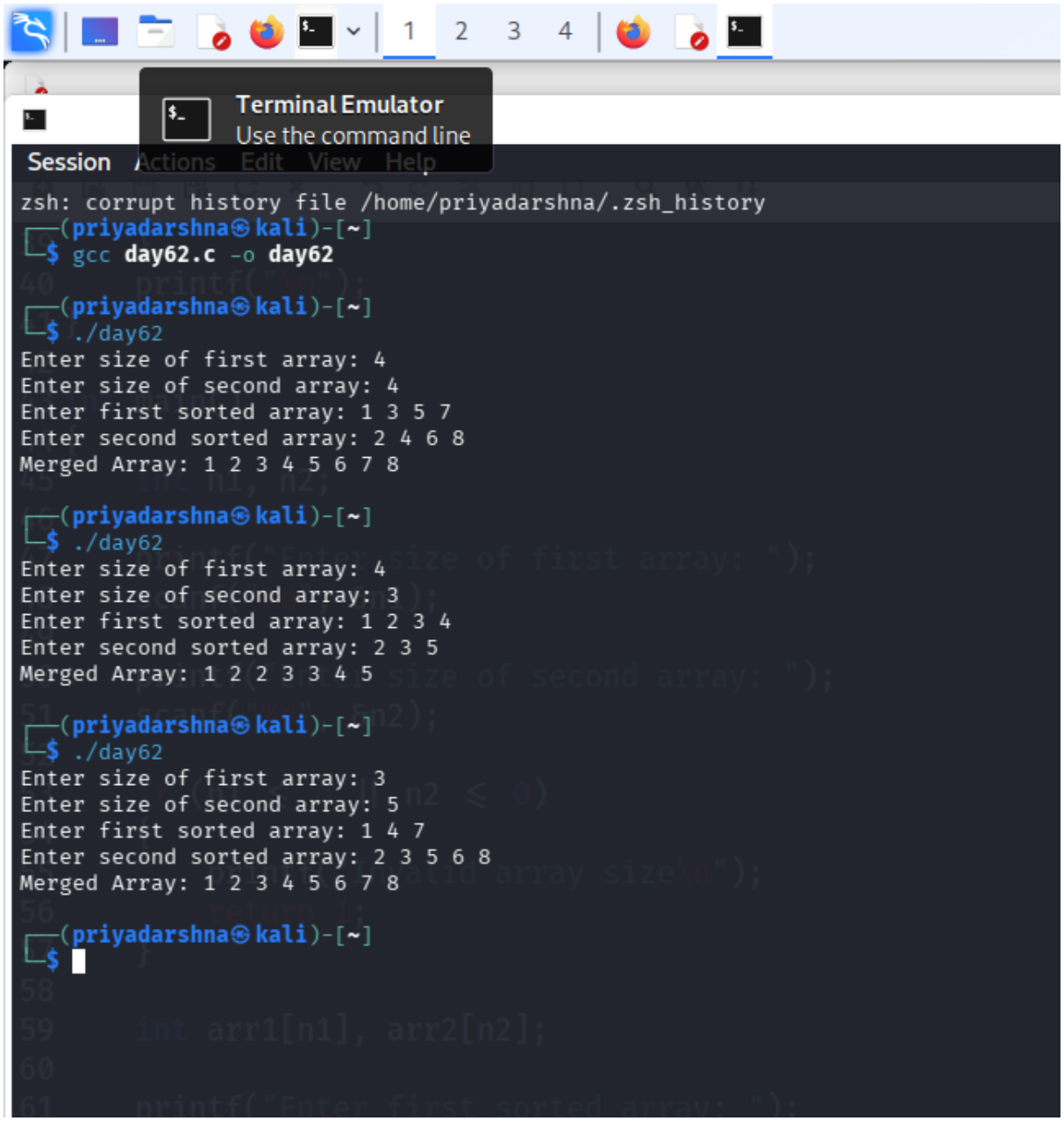
    printf("Enter first sorted array: ");
    for (int i = 0; i < n1; i++)
    {
        scanf("%d", &arr1[i]);
    }

    printf("Enter second sorted array: ");
    for (int i = 0; i < n2; i++)
    {
        scanf("%d", &arr2[i]);
    }

    merge_sorted_arrays(arr1, arr2, n1, n2);

    return 0;
}
```

>OUTPUT:


1 2 3 4
Terminal Emulator
Use the command line
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna㉿kali)-[~]
$ gcc day62.c -o day62
(priyadarshna㉿kali)-[~]
$ ./day62
Enter size of first array: 4
Enter size of second array: 4
Enter first sorted array: 1 3 5 7
Enter second sorted array: 2 4 6 8
Merged Array: 1 2 3 4 5 6 7 8
(priyadarshna㉿kali)-[~]
$ ./day62
Enter size of first array: 4
Enter size of second array: 3
Enter first sorted array: 1 2 3 4
Enter second sorted array: 2 3 5
Merged Array: 1 2 2 3 3 4 5
(priyadarshna㉿kali)-[~]
$ ./day62
Enter size of first array: 3
Enter size of second array: 5
Enter first sorted array: 1 4 7
Enter second sorted array: 2 3 5 6 8
Merged Array: 1 2 3 4 5 6 7 8
(priyadarshna㉿kali)-[~]
$